

BACKGROUND

- . Palbociclib PK variability: CV 45%
- . Standard 125 mg: 66% Grade 3/4
- . TDM targets Cmin < 100 ng/mL

OBJECTIVE

- Quantify clinical & economic benefit of TDM-guided dosing using PALOMA-calibrated population PK-PD modeling

METHODS

- Design: Monte Carlo (n=1,000)
- PK: Calibrated to PALOMA-2
- TDM: If Cmin > 100
- Reduce 125 -> 100 mg

Therapeutic Drug Monitoring of Palbociclib:
A PALOMA-Calibrated Population PK-PD Analysis

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TDM Prevents 158 Grade 3/4 Neutropenia Cases per 1,000 Patients
Number Needed to Treat (NNT) = 6.4 | Absolute Risk Reduction = 15.8%

Economic Benefit: ~\$3.4M Saved per 1,000 Patients

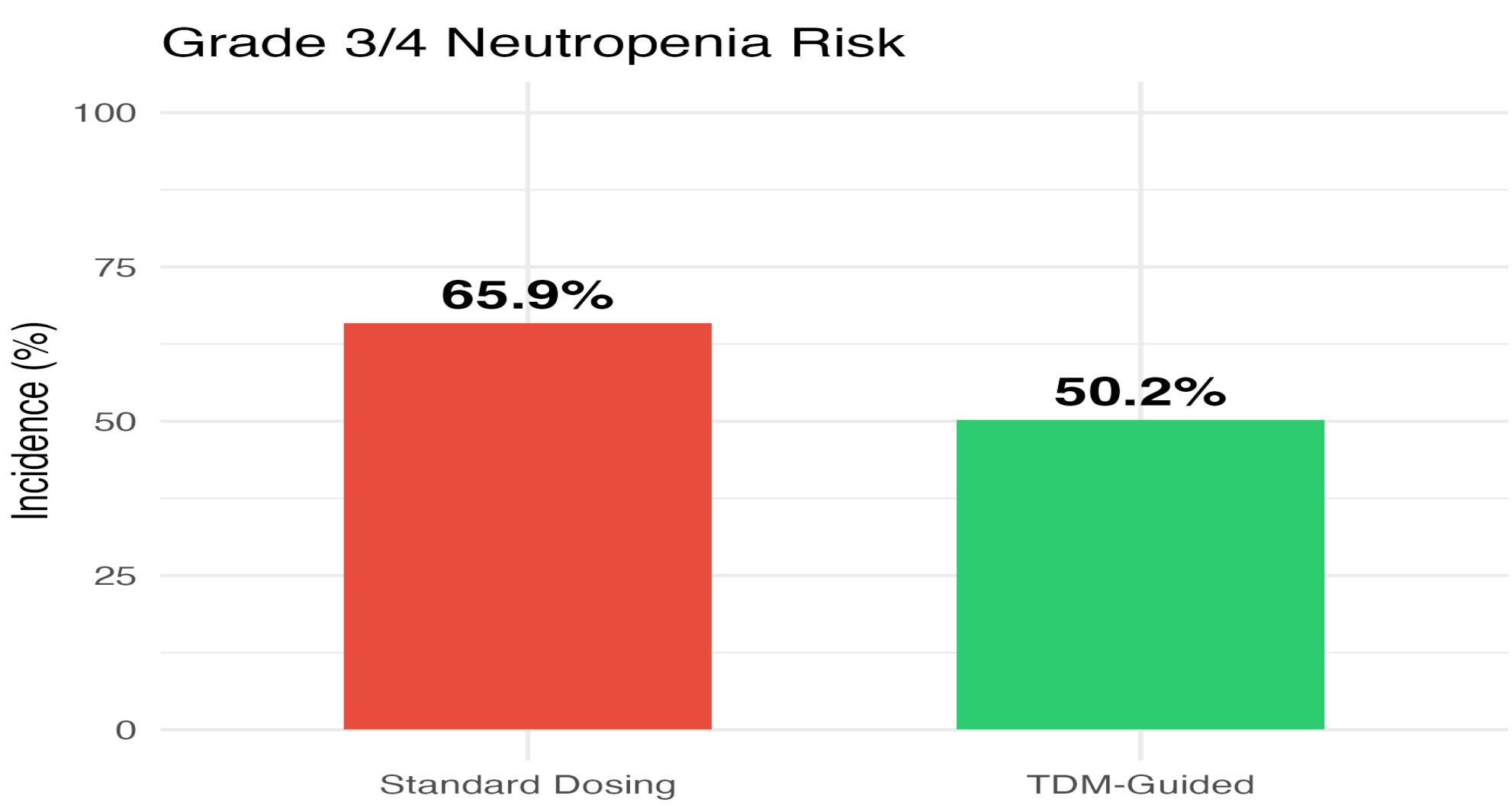


Fig 1: Risk Reduction

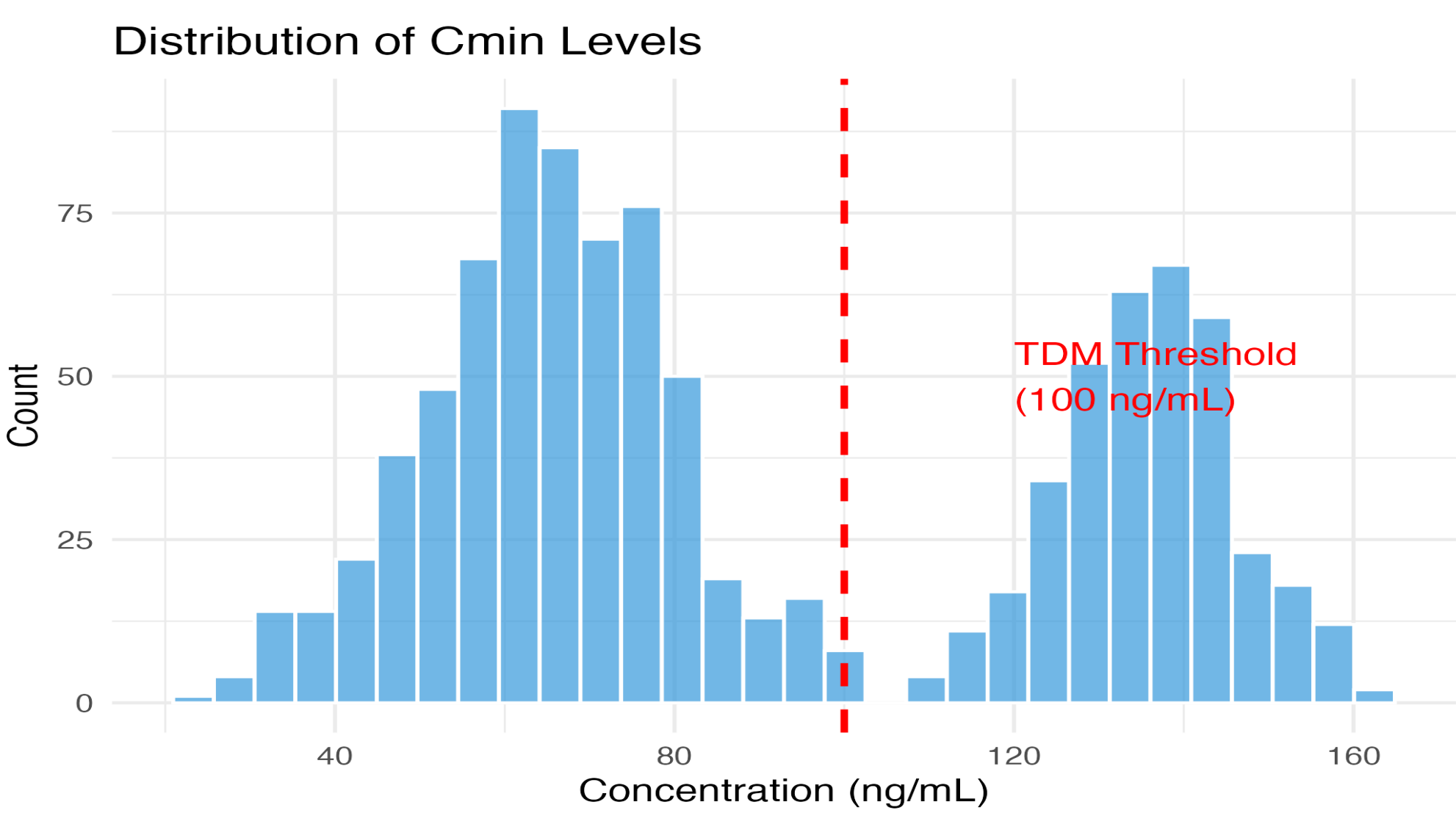


Fig 2: Cmin Distribution

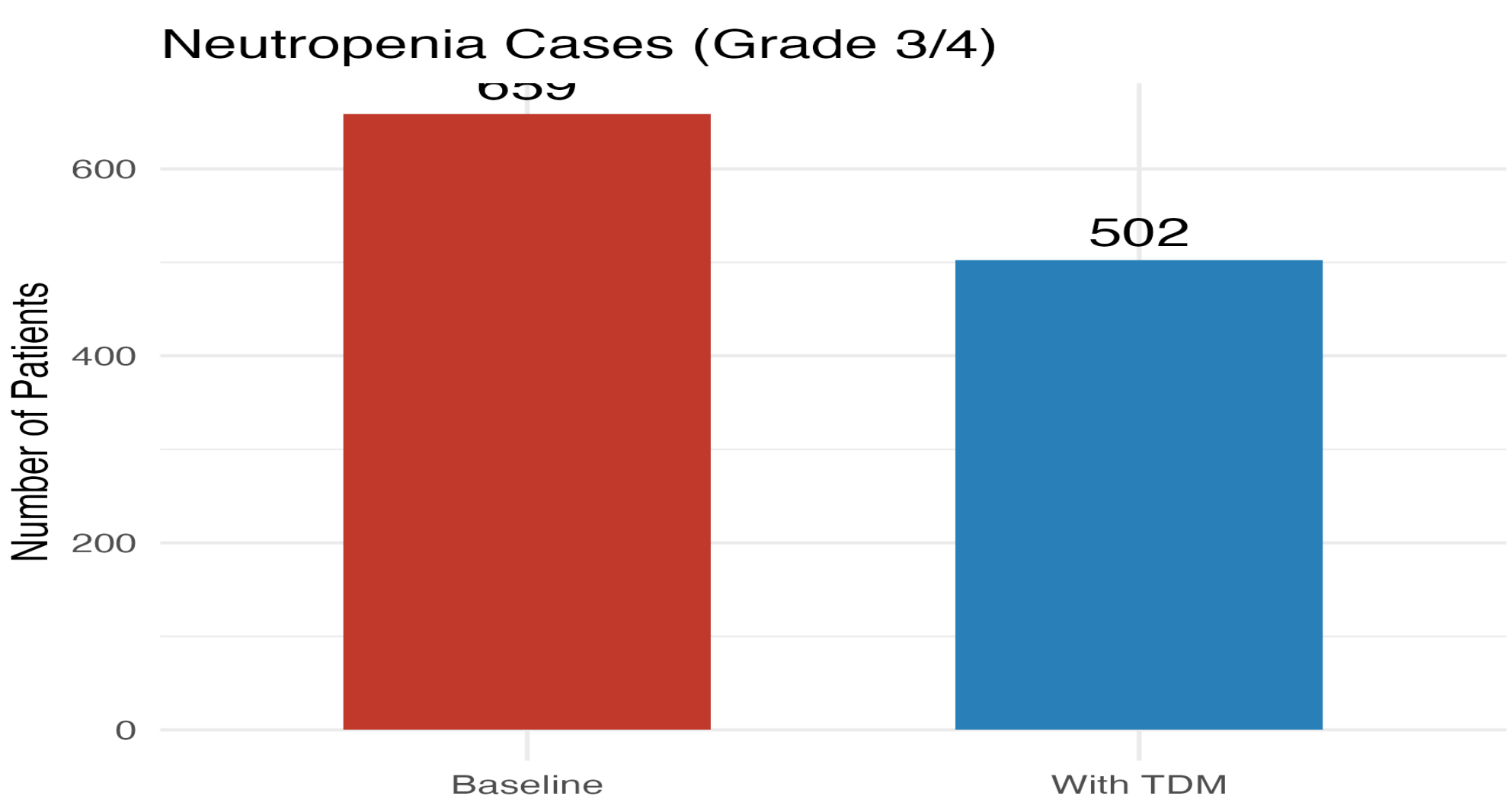


Fig 3: Cases Prevented

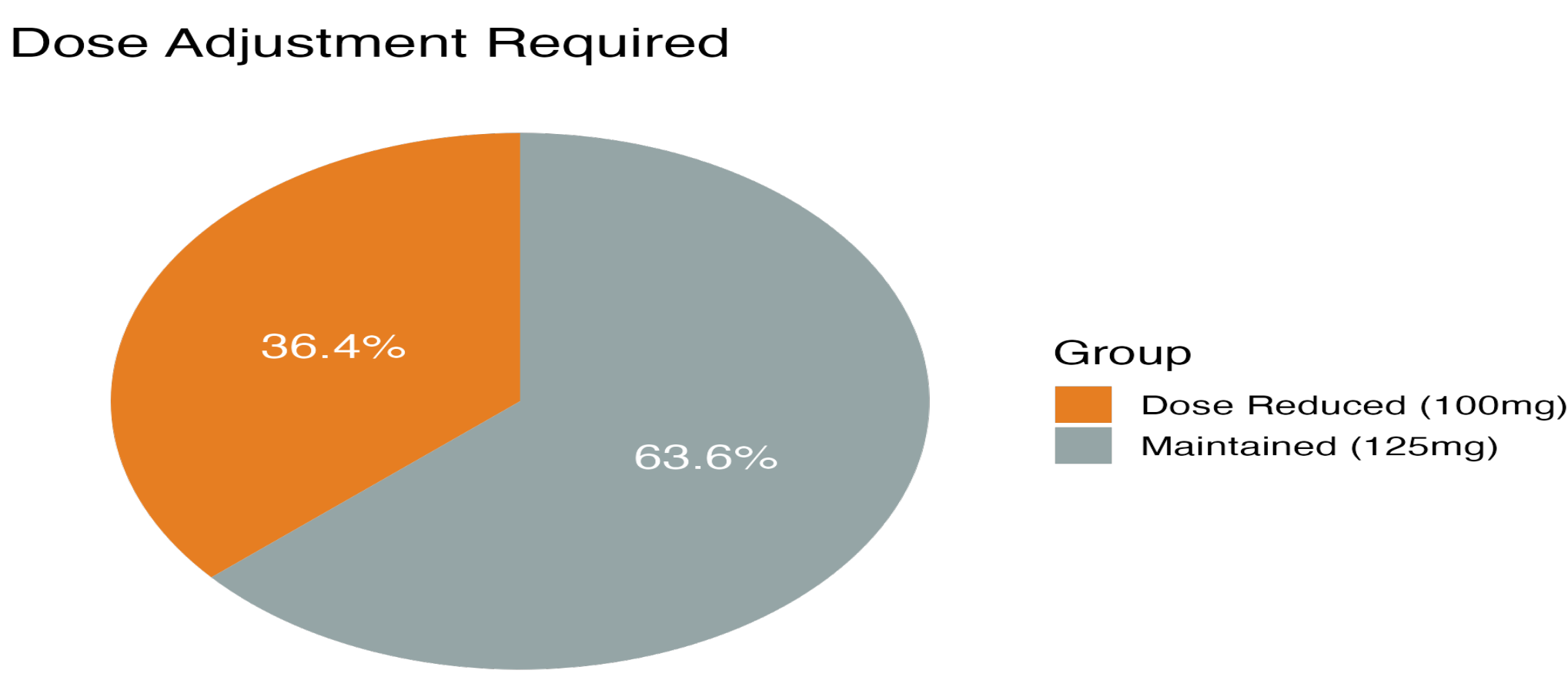


Fig 4: Dose Adjustments



Fig 5: NNT Visualization

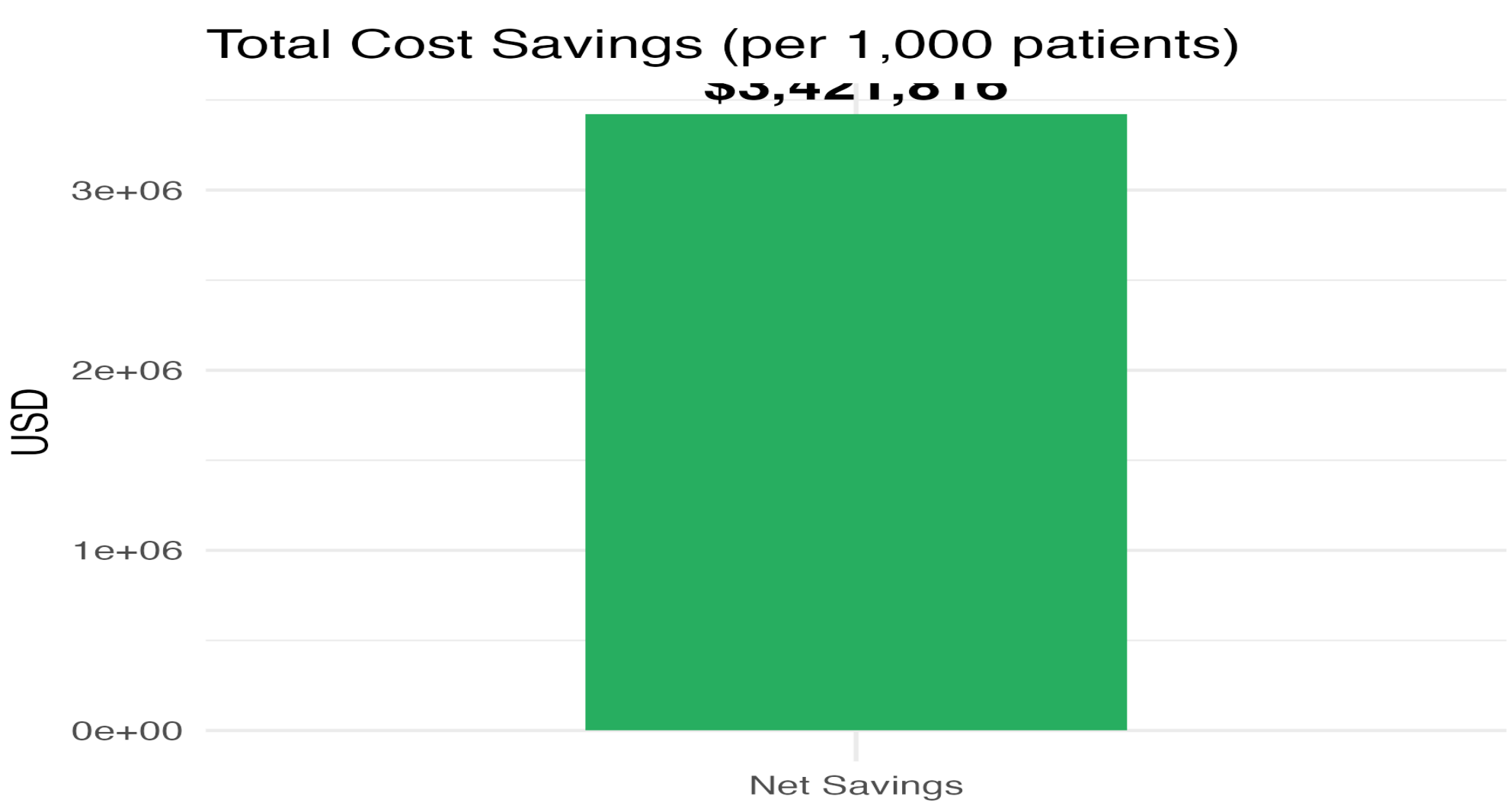


Fig 6: Economic Impact

RESULTS

- Baseline Risk: 65.9%
- TDM Risk: 50.2%
- Reduction: 15.8%
- NNT: 6.4 * EXCELLENT
- Cases Prevented: 158
- Dose Reduction: 36%

VALIDATION

- PALOMA-2 Alignment:
- . Baseline Risk: ~66%
- . Dose Red Rate: ~36%
- Robustness:
- . NNT stable (5.8-7.5)

CONCLUSION

- TDM effectively reduces severe neutropenia (NNT = 6.4) with major economic benefit (\$3.4M).